

Windows and WSL2

Using WSL2 on Windows

- This page explains how to use WSL2 (Windows Subsystem for Linux 2).
- All ASE 230 shell scripts and Linux command-line tools are written for Linux. **Windows users must run everything inside WSL2.**
- If you are using Linux or macOS, you can skip these slides.

1. Install WSL2 and Ubuntu

The easiest method works on Windows 11 and Windows 10 version 2004 or later.

1. Search for **PowerShell**, right-click it, and select **Run as administrator**.

2. Run this one command:

```
wsl --install
```

This enables WSL, installs WSL2, and installs Ubuntu by default. **Restart Windows** if prompted.

Official guide: <https://learn.microsoft.com/windows/wsl/install>

2. Finish the Ubuntu Setup

1. After restarting, open **Ubuntu** from the Start menu or Windows Terminal.
2. Wait while Ubuntu finishes installing.
3. Create a Linux username and password when prompted.

Nothing appears while you type a Linux password. This is normal - type it and press Enter.

Then update Ubuntu:

```
sudo apt update  
sudo apt upgrade -y
```

3. Verify WSL2 for HW1

Open **PowerShell on Windows** - not the Ubuntu terminal - and run:

```
wsl --list --verbose
```

Confirm that Ubuntu's **VERSION** column shows **2** .

If **wsl --install** only displays help, install Ubuntu explicitly:

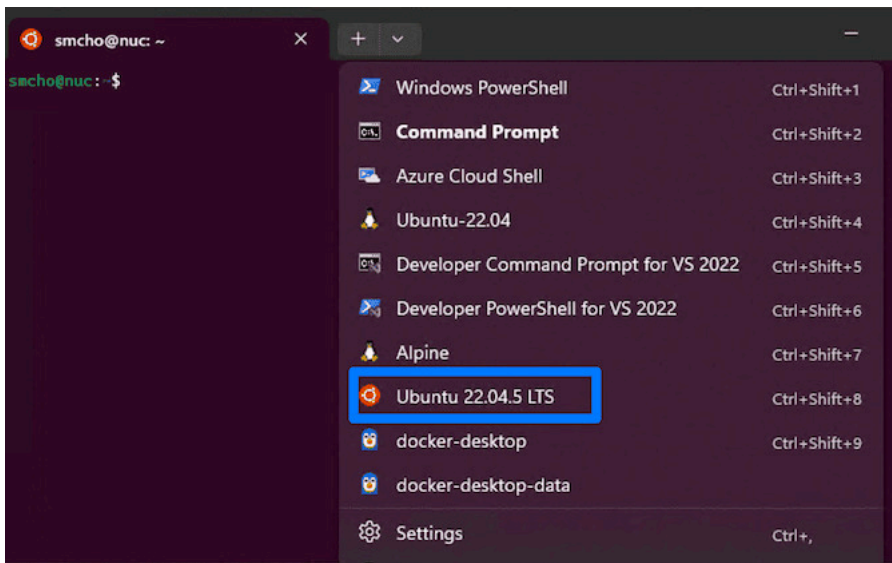
```
wsl --list --online  
wsl --install -d Ubuntu
```

If the download stays at **0.0%** , try:

```
wsl --install --web-download -d Ubuntu
```

4. Open the WSL2 (Ubuntu) Terminal

1. Open the Ubuntu terminal (new tab in Windows Terminal).



2. Do all course work - editing files and running scripts - inside WSL2.

5. Accessing Windows Files from WSL2

- The `C:` drive is mounted as `/mnt/c` inside WSL2.

```
# C:\temp\ex on Windows is the same as:  
cd /mnt/c/temp/ex
```

6. Line-Ending (EOL) Errors

Windows and Linux use different line endings. A script edited on Windows may fail like this:

```
./setup.sh: line 5: $'\r': command not found
```

Fix it with `dos2unix`:

```
sudo apt install dos2unix  
dos2unix ./setup.sh
```


Next: Install the Course Tools

After `wsl --list --verbose` confirms that Ubuntu is using WSL2, continue with:

1. `03. php installation.pdf`
2. `04. mysql installation.pdf`

Windows users should run the **Linux / WSL2 (Ubuntu)** commands in those guides inside the Ubuntu terminal.

Getting Unstuck

Configurations and repositories are updated frequently, so installation steps may occasionally cause errors. If that happens, you can:

- Copy and paste the error message into an LLM (Claude, ChatGPT, Perplexity, etc.) to get a quick fix.
- Email me the error details.
- Visit me in person so we can troubleshoot together.